Question No. 1: Electromagnetics (1/3)

2013年8月実施
問題1　電磁気学
（1頁目／3頁中）

Fig. 1に示すように，半無限の導体（導電率σ，誘電率ε，透磁率μ）と半無限の真空（誘電率ε_0，透磁率μ_0）が$x=0$のyz面で接している．導体中に角周波数ωで時間的に変化する電流がz方向に流れている．電流はy方向に一様分布であり，電流密度が$\boldsymbol{i}(x)$である．以下では，周波数が低く，変位電流を無視できる場合を考える．以下の問に答えよ．

(1) 周波数が低いときに導体中の伝導電流に比べ，変位電流を無視できる理由を説明せよ．

(2) 導体中の電流密度$\boldsymbol{i}(x)$を表す方程式

$$\nabla^2 \boldsymbol{i}(x) - j\omega\mu\sigma \boldsymbol{i}(x) = 0 \tag{1A}$$

をマクスウェル方程式から導出せよ．ただし，必要ならばベクトル演算の公式

$$\nabla \times \nabla \times \boldsymbol{A} = \nabla(\nabla \cdot \boldsymbol{A}) - \nabla^2 \boldsymbol{A} \tag{1B}$$

を用いてもよい．

(3) 式(1A)を解き，電流密度$\boldsymbol{i}(x)$の表示式を求めよ．ここで，導体表面$(x=0)$の電流密度を$\boldsymbol{i}_0$とする．

(4) 導体表面から深さX_0までの導体中に発生するy方向の単位幅，z方向の単位長あたりのジュール熱を求めよ．

(5) y方向の単位幅あたりの電流の大きさを一定としたとき，導体中に発生するジュール熱が周波数の平方根に比例することを証明せよ．

Question No. 1: Electromagnetics (2/3)

2013年8月実施
問題1　電磁気学
（2頁目／3頁中）

As shown in Fig. 1, a semi-infinite conductor (conductivity σ, permittivity ε, and permeability μ) and a semi-infinite vacuum (permittivity ε_0, permeability μ_0) are in contact at the yz plane at $x = 0$. A current varying with time with an angular frequency ω flows in the conductor along the z direction. The current is uniform along the y direction and the current density is $\boldsymbol{i}(x)$. In the following, it is assumed that the frequency is so low that the displacement current can be ignored. Answer the following questions.

(1) Explain why the displacement current can be ignored compared with the conduction current when the frequency is low.

(2) By using Maxwell's equations, derive the following equation which expresses the current density $\boldsymbol{i}(x)$ in the conductor,

$$\nabla^2 \boldsymbol{i}(x) - j\omega\mu\sigma \boldsymbol{i}(x) = 0 \tag{1A}$$

If necessary, use the vector formula

$$\nabla \times \nabla \times \boldsymbol{A} = \nabla(\nabla \cdot \boldsymbol{A}) - \nabla^2 \boldsymbol{A} \tag{1B}$$

(3) Solve equation (1A) and derive an expression for the current density $\boldsymbol{i}(x)$. Here, the current density on the conductor surface ($x = 0$) is assumed to be $\boldsymbol{i}_0$.

(4) Derive the Joule heat in the conductor from the surface of the conductor to a depth of X_0, per unit width in the y direction and per unit length in the z direction.

(5) Show that the Joule heat in the conductor is proportional to the square root of frequency when the magnitude of the current in the conductor per unit width in y direction is constant.

Question No. 1: Electromagnetics (3/3)

2013年8月実施
問題１　電磁気学
（３頁目／３頁中）

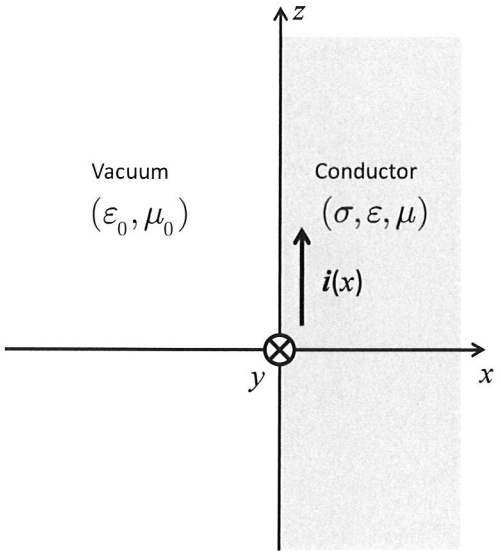


Fig. 1

Question No. 2: Electrical circuits (1/3)

2013 年 8 月実施
問題 2　電気回路
(1 頁目／3 頁中)

Fig. 2 (a) に示す未知の二端子対網 N について以下の問に答えよ．ここで，電圧 V_1, V_2 と電流 I_1, I_2 の関係はインピーダンスパラメータ $(z_{11}, z_{12}, z_{21}, z_{22})$ を用いて次式のように与えられる．

$$V_1 = z_{11}I_1 + z_{12}I_2 \qquad V_2 = z_{21}I_1 + z_{22}I_2$$

(1) 以下の開放電流および電圧は，二端子対網 N に対し実験的に決定された．

$$\left.\begin{aligned} V_1 &= 10\angle 0°\text{[V]} \\ V_2 &= 6\angle 90°\text{[V]} \\ I_1 &= 1\angle 0°\text{[A]} \end{aligned}\right|_{I_2=0\text{[A]}} \qquad \left.\begin{aligned} V_1 &= 15\angle 90°\text{[V]} \\ V_2 &= 10\angle 0°\text{[V]} \\ I_2 &= 2.5\angle 0°\text{[A]} \end{aligned}\right|_{I_1=0\text{[A]}}$$

フェーザ表示 $r\angle\theta$ において，r と θ はそれぞれ電流や電圧の実効値および初期位相である．インピーダンスパラメータ $(z_{11}, z_{12}, z_{21}, z_{22})$ を決定せよ．

(2) 二端子対網 N の T 型等価回路を Fig. 2 (b)に示す．インピーダンス Z_1，Z_2，Z_3 を求めよ．

(3) Fig. 2 (c) に示すように，二端子対網 N に交流電圧源（振幅 15[V]，角周波数 3.2[rad/s]，内部抵抗 5[Ω]）と，コイル（インダクタンス 2[H]）を接続する．端子 2-2'から左側についてテブナンの等価電源（電圧源）を求め，電圧 $v_0(t)$を計算せよ．

Question No. 2: Electrical circuits (2/3)

2013 年 8 月実施
問題 2　電気回路
(2 頁目／3 頁中)

Answer the following questions about the unknown two-port network N shown in Fig. 2 (a). The voltages V_1, V_2 and currents I_1, I_2 can be related in terms of impedance parameters (z_{11}, z_{12}, z_{21}, z_{22}) as shown below.

$$V_1 = z_{11}I_1 + z_{12}I_2 \qquad V_2 = z_{21}I_1 + z_{22}I_2$$

(1) The following open-circuit currents and voltages were determined experimentally for the two-port network N:

$$\left.\begin{aligned} V_1 &= 10\angle 0°\text{[V]} \\ V_2 &= 6\angle 90°\text{[V]} \\ I_1 &= 1\angle 0°\text{[A]} \end{aligned}\right|_{I_2=0\text{[A]}} \qquad \left.\begin{aligned} V_1 &= 15\angle 90°\text{[V]} \\ V_2 &= 10\angle 0°\text{[V]} \\ I_2 &= 2.5\angle 0°\text{[A]} \end{aligned}\right|_{I_1=0\text{[A]}}$$

In the phasor equivalent $r\angle\theta$, r and θ denote the rms value and the initial phase of a voltage or a current, respectively. Determine the impedance parameters (z_{11}, z_{12}, z_{21}, z_{22}).

(2) The T-equivalent circuit for the two-port network N is shown in Fig. 2 (b). Determine the impedance Z_1, Z_2 and Z_3.

(3) An ac voltage source (amplitude 15[V], angular frequency 3.2[rad/s], internal resistance 5[Ω]) and a coil (inductance 2[H]) are connected to the two-port network N as shown in Fig. 2 (c). Find the Thevenin equivalent (voltage source) to the left of terminals 2-2'and calculate the voltage $v_0(t)$.

Question No. 2: Electrical circuits (3/3)

2013 年 8 月実施
問題 2　電気回路
(3 頁目／3 頁中)

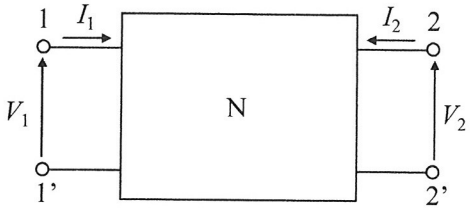


Fig. 2 (a)

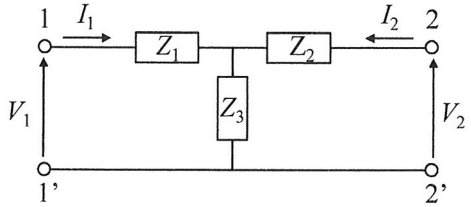


Fig. 2 (b)

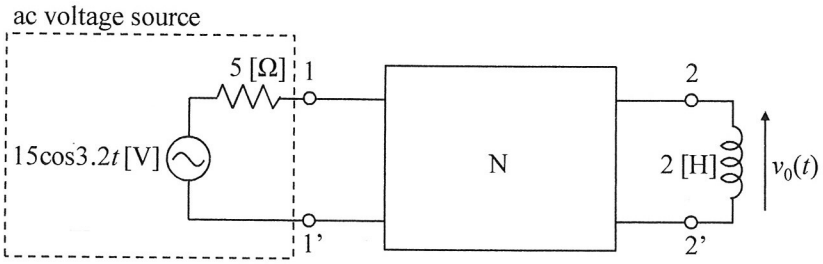


Fig. 2 (c)

Question No. 3: Basic information science 1 (1/1)

2013 年 8 月実施
問題 3　情報基礎 1
(1 頁目／1 頁中)

重みと呼ばれる実数ベクトル $(a_1, a_2, \ldots, a_n)$ としきい値と呼ばれる実数 b が存在し，n 変数論理関数 $f(x_1, x_2, \ldots, x_n)$ が

$$f(x_1, x_2, \ldots, x_n) = \begin{cases} 1 & a_1x_1 + a_2x_2 + \cdots + a_nx_n \geq b \text{ のとき} \\ 0 & \text{その他} \end{cases}$$

を満たすとき，f をしきい値関数という．以下の問に答えよ．なお，$\wedge$ は論理積演算，$\vee$ は論理和演算，$\overline{\ }$ は否定演算を表すとする．$x_1, x_2, \ldots, x_n \in \{0,1\}$ とする．

(1) $a_1 = 5,\ a_2 = 4,\ a_3 = 3,\ a_4 = 2,\ a_5 = 1,\ b = 8$ が与えられているとき，5 変数しきい値関数 f を否定演算を含まない積和形の論理式で表せ．

(2) 次の関数は $a_1 = a_2 = a_3 = 1$ である 3 変数しきい値関数である．それぞれにおけるしきい値 b の値を求めよ．

(a) $f_1(x_1, x_2, x_3) = x_1 \wedge x_2 \wedge x_3$

(b) $f_2(x_1, x_2, x_3) = (x_1 \wedge x_2) \vee (x_2 \wedge x_3) \vee (x_3 \wedge x_1)$

(3) 同じ実数ベクトルを重みとして持つ 2 つの n 変数しきい値関数 f_1 と f_2 において，それぞれのしきい値 b_1 と b_2 の間に $b_1 \geq b_2$ の関係があるとき，任意の $x_1, x_2, \ldots, x_n$ に対して $f_1(x_1, x_2, \ldots, x_n) \leq f_2(x_1, x_2, \ldots, x_n)$ が成立するか否かを判定し，その根拠を示せ．

(4) $f(x_1, x_2, \ldots, x_n)$ を $a_1 = a_2 = \cdots = a_n = 1$ かつ b が $0 \leq b \leq n$ の整数である n 変数しきい値関数とする．

(a) $\overline{f(\overline{x_1}, \overline{x_2}, \ldots, \overline{x_n})}$ がしきい値関数であることを示せ．

(b) $f(x_1, x_2, \ldots, x_n) = \overline{f(\overline{x_1}, \overline{x_2}, \ldots, \overline{x_n})}$ を満たすための n と b に関する必要十分条件を示せ．

With the existences of a real vector $(a_1, a_2, \ldots, a_n)$ as weight and a real number b as threshold, an n-variable Boolean function $f(x_1, x_2, \ldots, x_n)$ is called a threshold function if and only if it satisfies

$$f(x_1, x_2, \ldots, x_n) = \begin{cases} 1 & \text{if } a_1x_1 + a_2x_2 + \cdots + a_nx_n \geq b, \\ 0 & \text{otherwise.} \end{cases}$$

Answer the following questions. Denote AND, OR, and NOT operators by $\wedge$, $\vee$, and $\overline{\ }$, respectively. Consider $x_1, x_2, \ldots, x_n \in \{0,1\}$.

(1) Find a sum-of-products expression without NOT operators for the Boolean expression of a 5-variable threshold function f when $a_1 = 5,\ a_2 = 4,\ a_3 = 3,\ a_4 = 2,\ a_5 = 1$, and $b = 8$.

(2) The following functions are 3-variable threshold functions with $a_1 = a_2 = a_3 = 1$. Find the value of the threshold b in each function.

(a) $f_1(x_1, x_2, x_3) = x_1 \wedge x_2 \wedge x_3$

(b) $f_2(x_1, x_2, x_3) = (x_1 \wedge x_2) \vee (x_2 \wedge x_3) \vee (x_3 \wedge x_1)$

(3) Two n-variable threshold functions f_1 and f_2 have the same weight vector and different thresholds b_1 and b_2 $(b_1 \geq b_2)$, respectively. Determine whether $f_1(x_1, x_2, \ldots, x_n) \leq f_2(x_1, x_2, \ldots, x_n)$ is satisfied or not for any $x_1, x_2, \ldots, x_n$, and justify your answer.

(4) Let $f(x_1, x_2, \ldots, x_n)$ be an n-variable threshold function where $a_1 = a_2 = \cdots = a_n = 1$ and b is an integer satisfying $0 \leq b \leq n$.

(a) Prove that $\overline{f(\overline{x_1}, \overline{x_2}, \ldots, \overline{x_n})}$ is a threshold function.

(b) Find the necessary and sufficient condition in terms of n and b to satisfy $f(x_1, x_2, \ldots, x_n) = \overline{f(\overline{x_1}, \overline{x_2}, \ldots, \overline{x_n})}$.

Question No. 4: Basic information science 2 (1/1)

2013年 8月実施
問題4 情報基礎2
(1頁目／1頁中)

配列 A には n 個の相異なる整数 $A[1], A[2], \ldots, A[n]$ が含まれているものとする．次の問に答えよ．

(1) 配列 A と整数 x が入力として与えられたとき，A の要素と x との差の最小値，すなわち

$$\min\{\ |A[i] - x|\ :\ 1 \leq i \leq n\ \}$$

を計算する問題 P を考える．

(a) P を $O(n)$ 時間で解くアルゴリズムの概略を示せ．

(b) 配列 A の要素があらかじめ昇順に並んでいると仮定したとき，P をより効率的に解くアルゴリズムの概略を示し，その時間計算量を与えよ．

(2) 配列 A が入力として与えられたとき，A の二つの要素の差の**最小値**，すなわち

$$\min\{\ |A[i] - A[j]|\ :\ 1 \leq i, j \leq n,\quad i \neq j\ \}$$

を計算する問題 Q を考える．Q を解く $O(n \log n)$ 時間のアルゴリズムの概略を示せ．

(3) 配列 A が入力として与えられたとき，A の二つの要素の差の**最大値**，すなわち

$$\max\{\ |A[i] - A[j]|\ :\ 1 \leq i, j \leq n,\quad i \neq j\ \}$$

を計算する問題 R を考える．R を効率よく解くアルゴリズムの概略を示し，その時間計算量を与えよ．

We assume that an array A contains n distinct integers $A[1], A[2], \ldots, A[n]$. Answer the following questions.

(1) Given an array A and an integer x as input, we consider the problem P to compute the value

$$\min\{\ |A[i] - x|\ :\ 1 \leq i \leq n\ \},$$

that is the smallest difference between an element in A and x.

(a) Outline an algorithm that solves P in $O(n)$ time.

(b) Outline a more efficient algorithm to solve P and give its time complexity, assuming that the array A is sorted in increasing order beforehand.

(2) Given an array A as input, we consider the problem Q to compute the value

$$\min\{\ |A[i] - A[j]|\ :\ 1 \leq i, j \leq n,\quad i \neq j\ \},$$

that is the *smallest* difference between two elements in A. Outline an algorithm that solves Q in $O(n \log n)$ time.

(3) Given an array A as input, we consider the problem R to compute the value

$$\max\{\ |A[i] - A[j]|\ :\ 1 \leq i, j \leq n,\quad i \neq j\ \},$$

that is the *largest* difference between two elements in A. Outline an efficient algorithm to solve R and give its time complexity.

Question No. 5: Basic physics 1 (1/3)

2013 年 8 月実施
問題 5　物理基礎 1
(1 頁目／3 頁中)

Fig. 5 に示すように，一端が支点 **P** に固定された棒からなる振り子を考える．棒は鉛直面内で自由に回転できる．棒の質量と長さをそれぞれM，$2l$とする．棒と支点を通る水平軸のなす角度をθとする．棒の太さは無視できその密度は一様であると仮定する．また空気抵抗や振り子の支点 **P** での摩擦はないものと仮定する．以下の問に答えよ．

(1) 棒の重心を通り棒の長さ方向に垂直な軸のまわりの慣性モーメントが $\frac{1}{3}Ml^2$ で与えられることを示せ．また振り子の支点のまわりの棒の慣性モーメントを求めよ．

(2) 鉛直下方（$\theta = \pi/2$）の位置から，ある初期角速度を与えて回転させたい．回転させるのに必要な最小の初期角速度を求めよ．

(3) 振り子の回転の運動方程式を求めよ．

(4) 棒を水平な位置（$\theta = 0$）から放した．

(a) 棒を放した後の任意の位置での角速度を求めよ．

必要に応じて $\frac{d}{dt}\dot{\theta}^2 = 2\dot{\theta}\cdot\ddot{\theta}$ を用いよ．

(b) 支点での抗力の水平方向成分と鉛直方向成分を求めよ．

Question No. 5: Basic physics 1 (2/3)

2013 年 8 月実施
問題 5　物理基礎 1
(2 頁目／3 頁中)

As shown in Fig. 5, consider a pendulum that consists of a rod one end of which is fixed at a pivot point **P**. The rod can rotate freely in a vertical plane. Let the mass and length of the rod be M and $2l$, respectively. Let the angle between the rod and the horizontal axis through the pivot point be θ. Assume that the width of the rod is negligible and the density of the rod is uniform. In addition, assume that there is no air resistance and no friction at the pivot point **P**. Answer the following questions.

(1) Show that the moment of inertia of the rod about the axis through the center of mass perpendicular to the length direction of the rod is given by $\frac{1}{3}Ml^2$. Then, find the moment of inertia of the rod around the pivot point of the pendulum.

(2) Let us give a certain initial angular velocity to a pendulum at rest in the vertical position ($\theta = \pi/2$), causing the pendulum to rotate around the pivot point. Derive the minimum initial angular velocity required to cause rotation.

(3) Find the equation of motion for the rotation of the pendulum.

(4) The rod was released from the horizontal position ($\theta = 0$).

(a) Find the angular velocity at an arbitrary position after the rod was released.

Use $\frac{d}{dt}\dot{\theta}^2 = 2\dot{\theta} \cdot \ddot{\theta}$, if necessary.

(b) Derive the vertical and horizontal components of the reaction for the pivot point.

Question No. 5: Basic physics 1 (3/3)

2013 年 8 月実施
問題 5　物理基礎 1
(3 頁目／3 頁中)

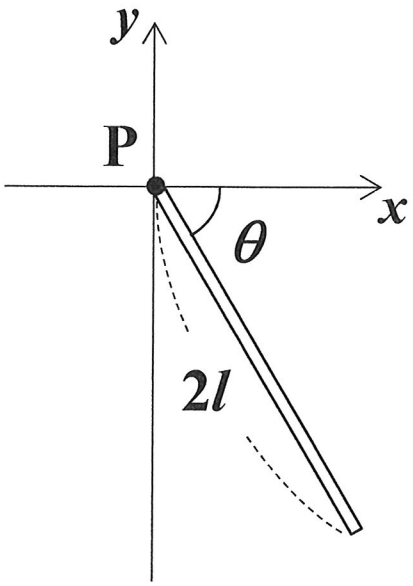


Fig. 5

Question No. 6: Basic physics 2 (1/2)

2013年8月実施
問題6　物理基礎2
（1頁目／2頁中）

3次元座標空間における2次曲面を表す式

$$25(a-1)x^2+(9b+16)y^2+(16b+9)z^2-24(b-1)yz=25 \quad \cdots (6A)$$

と平面を表す式

$$x+\frac{4}{5}y+\frac{3}{5}z=1 \quad \cdots (6B)$$

について考える．ここで$a>2$，$b<0$とする．以下の問に答えよ．

(1) 次の関係式を満たす3行3列の対称行列$\mathbf{F}$を求めよ．

$$25(a-1)x^2+(9b+16)y^2+(16b+9)z^2-24(b-1)yz=\begin{pmatrix}x & y & z\end{pmatrix}\mathbf{F}\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

(2) $\mathbf{F}$の全ての固有値λ_1，λ_2，λ_3（$\lambda_1>\lambda_2>\lambda_3$）とこれらに対応する規格化された固有ベクトルを求めよ．次の関係式を満たす1つの直交行列$\mathbf{P}$とその逆行列$\mathbf{P}^{-1}$を求めよ．

$$\mathbf{P}^{-1}\mathbf{F}\mathbf{P}=\begin{pmatrix}\lambda_1 & 0 & 0\\ 0 & \lambda_2 & 0\\ 0 & 0 & \lambda_3\end{pmatrix}$$

(3) 式(6B)を直交行列$\mathbf{P}$によって変換された新しい座標変数x'，y'，z'を用いて表せ．

(4) 2次曲面(式(6A))と平面(式(6B))の交点を表す曲線の式を直交行列$\mathbf{P}$によって変換された新しい座標変数x'，y'，z'を用いて表せ．交点の曲線の概略を図示せよ．

Question No. 6: Basic physics 2 (2/2)

2013年8月実施
問題6　物理基礎2
（2頁目／2頁中）

Consider the equation

$$25(a-1)x^2+(9b+16)y^2+(16b+9)z^2-24(b-1)yz=25 \quad \cdots \text{(6A)}$$

which represents a quadratic plane and the equation

$$x+\frac{4}{5}y+\frac{3}{5}z=1 \quad \cdots \text{(6B)}$$

which represents a flat plane in three-dimensional coordinate space. Here, $a>2$ and $b<0$. Answer the following questions.

(1) Find a 3×3 symmetrical matrix $\mathbf{F}$ which satisfies the following equation

$$25(a-1)x^2+(9b+16)y^2+(16b+9)z^2-24(b-1)yz=\begin{pmatrix}x & y & z\end{pmatrix}\mathbf{F}\begin{pmatrix}x\\ y\\ z\end{pmatrix}.$$

(2) Find all the eigenvalues λ_1, λ_2 and λ_3 ($\lambda_1>\lambda_2>\lambda_3$) of $\mathbf{F}$ and their corresponding normalized eigenvectors. Find an orthogonal matrix $\mathbf{P}$ and its inverse matrix $\mathbf{P}^{-1}$ that satisfy the following equation:

$$\mathbf{P}^{-1}\mathbf{F}\mathbf{P}=\begin{pmatrix}\lambda_1 & 0 & 0\\ 0 & \lambda_2 & 0\\ 0 & 0 & \lambda_3\end{pmatrix}.$$

(3) Express equation (6B) using new coordinate variables x', y' and z' which are transformed by the orthogonal matrix $\mathbf{P}$.

(4) Find an equation which represents the intersection curve between the quadratic plane (equation (6A)) and the flat plane (equation (6B)) using the new coordinate variables x', y' and z' which are transformed by the orthogonal matrix $\mathbf{P}$. Sketch the outline of the intersection curve.